Project One Template

MAT350: Applied Linear Algebra

Kristi Huggins

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Problem 1

Develop a system of linear equations for the network by writing an equation for each router (A, B, C, D, and E). Make sure to write your final answer as A**x**=**b** where A is the 5x5 coefficient matrix, **x** is the 5x1 vector of unknowns, and **b** is a 5x1 vector of constants.

Solution:

Router A:

$x_1 + 2x_2 + 0x_3 + 0x_4 + 0x_5 = 100$

Router B:

$x_1 + x_2 - x_3 + 0x_4 - x_5 = 0$

Router C:

$0x_1 + x_2 - x_3 + 0x_4 - x_5 = -50$

Router D:

$0x_1 - x_2 + 0x_3 + x_4 + x_5 = 120$

Router E:

$0x_1 + x_2 + x_3 - x_4 + x_5 = 0$

Problem 2

Use MATLAB to construct the augmented matrix [A **b**] and then perform row reduction using the rref() function. Write out your **reduced matrix and identify the free and basic variables of the system**.

Solution:

```
%code
A = [1 2 0 0 0; 1 1 -1 0 -1; 0 1 -1 0 -1; 0 -1 0 1 1; 0 1 1 -1 1]
```

```
A = 5×5
     1     2     0     0     0
     1     1    -1     0    -1
```

```
     0     1    -1     0    -1
     0    -1     0     1     1
     0     1     1    -1     1
```

```
b = [100; 0; -50; 120; 0]
```

```
b = 5×1
   100
     0
   -50
   120
     0
```

```
Ab = [A b]
```

```
Ab = 5×6
     1     2     0     0     0   100
     1     1    -1     0    -1     0
     0     1    -1     0    -1   -50
     0    -1     0     1     1   120
     0     1     1    -1     1     0
```

```
[rowreducedAugAb, pivotvarsAugAb] = rref(Ab)
```

```
rowreducedAugAb = 5×6
     1     0     0     0     0    50
     0     1     0     0     0    25
     0     0     1     0     0    30
     0     0     0     1     0   100
     0     0     0     0     1    45
pivotvarsAugAb = 1×5
     1     2     3     4     5
```

```
[numeqns, numvars] = size(A)
```

```
numeqns =
5
numvars =
5
```

```
numpivotvars = length(pivotvarsAb)
```

```
numpivotvars =
5
```

```
numfreevars = numvars - numpivotvars
```

```
numfreevars =
0
```

Problem 3

Use MATLAB to **compute the LU decomposition of A**, i.e., find A = LU. For this decomposition, find the transformed set of equations L**y** = **b**, where **y** = U**x**. Solve the system of equations L**y** = **b** for the unknown vector **y**.

Solution:

```
%code
[L, U] = lu(A)
```

```
L = 5×5
    1.0000         0         0         0         0
    1.0000    1.0000         0         0         0
         0   -1.0000    1.0000         0         0
         0    1.0000   -0.5000    1.0000         0
         0   -1.0000         0   -1.0000    1.0000
U = 5×5
     1     2     0     0     0
     0    -1    -1     0    -1
     0     0    -2     0    -2
     0     0     0     1     1
     0     0     0     0     1
```

```
y = L\b
```

```
y = 5×1
   100
  -100
  -150
   145
    45
```

```
x = U\y
```

```
x = 5×1
    50
    25
    30
   100
    45
```

Problem 4

Use MATLAB to **compute the inverse** of U using the inv() function.

Solution:

```
%code
invU = inv(U)
```

```
invU = 5×5
    1.0000    2.0000   -1.0000         0         0
         0   -1.0000    0.5000         0         0
         0         0   -0.5000         0   -1.0000
         0         0         0    1.0000   -1.0000
```

```
         0         0         0         0    1.0000
```

Problem 5

Compute the solution to the original system of equations by transforming **y** into **x**, i.e., compute **x** = inv(U)**y**.

Solution:

```
%code
x = inv(U)*y
```

```
x = 5×1
    50
    25
    30
   100
    45
```

Problem 6

Check your answer for x_1 using Cramer's Rule. Use MATLAB to compute the required determinants using the det() function.

Solution:

```
%code
detA = det(A)
```

```
detA =
2
```

```
A1 = A; A1(:,1) = b
```

```
A1 = 5×5
   100     2     0     0     0
     0     1    -1     0    -1
   -50     1    -1     0    -1
   120    -1     0     1     1
     0     1     1    -1     1
```

```
A2 = A; A2(:,2) = b
```

```
A2 = 5×5
     1   100     0     0     0
     1     0    -1     0    -1
     0   -50    -1     0    -1
     0   120     0     1     1
     0     0     1    -1     1
```

```
A3 = A; A3(:,3) = b
```

```
A3 = 5×5
     1     2   100     0     0
     1     1     0     0    -1
     0     1   -50     0    -1
     0    -1   120     1     1
     0     1     0    -1     1
```

```
A4 = A; A4(:,4) = b
```

```
A4 = 5×5
     1     2     0   100     0
     1     1    -1     0    -1
     0     1    -1   -50    -1
     0    -1     0   120     1
     0     1     1     0     1
```

```
A5 = A; A5(:,5) = b
```

```
A5 = 5×5
     1     2     0     0   100
     1     1    -1     0     0
     0     1    -1     0   -50
     0    -1     0     1   120
     0     1     1    -1     0
```

```
detA1 = det(A1)
```

```
detA1 =
100
```

```
detA2 = det(A2)
```

```
detA2 =
50
```

```
detA3 = det(A3)
```

```
detA3 =
60.0000
```

```
detA4 = det(A4)
```

```
detA4 =
200
```

```
detA5 = det(A5)
```

```
detA5 =
90
```

```
x1_cramer = detA1 / detA
```

```
x1_cramer =
50
```

```
x2_cramer = detA2 / detA
```

```
x2_cramer =
25
```

```
x3_cramer = detA3 / detA
```

```
x3_cramer =
30.0000
```

```
x4_cramer = detA4 / detA
```

```
x4_cramer =
100
```

```
x5_cramer = detA5 / detA
```

```
x5_cramer =
45
```

Problem 7

The Project One Table Template, provided in the Project One Supporting Materials section in Brightspace, shows the recommended throughput capacity of each link in the network. Put your solution for the system of equations in the third column so it can be easily compared to the maximum capacity in the second column. In the fourth column of the table, provide recommendations for how the network should be modified based on your network throughput analysis findings. The modification options can be No Change, Remove Link, or Upgrade Link. In the final column, explain how you arrived at your recommendation.

Solution:

Network Link	Recommended Capacity (Mbps)	Solution	Recommendation	Explanation
x_1	60	0	Remove Link	The determinant-based solution shows zero flow on this link. Since it carries no traffic, it does not contribute to throughput and can be removed unless redundancy is required.
x_2	50	100	Upgrade Link	The computed flow (100 Mbps) exceeds the recommended capacity (50 Mbps). This indicates the link is overloaded, so upgrading is necessary to prevent congestion.
x_3	100	50	No Change	The actual flow is well below the recommended capacity. The link is performing within acceptable limits and does not require modification.
x_4	100	120	Upgrade Link	The flow exceeds the recommended capacity. This link is at risk of becoming a bottleneck, so upgrading it will ensure reliable throughput.
x_5	50	30	No Change	The flow is comfortably below capacity. The link is functioning efficiently and does not need modification.